EXECUTIVE SUMMARY

Intelligent Mobility: AI Meets Real Transportation

A companion to the full symposium proceedings · Rhode Island Department of Transportation

DATES
August 19–20, 2026
Tuesday & Wednesday

LOCATION
New England Institute of Technology
West Greenwich, Rhode Island

PARTICIPANTS
Federal, state, global, academic, industry
Keynotes · panels · peer exchange

AT A GLANCE

Where AI meets the real work of transportation

Over two days, RIDOT convened more than 100 leaders and technical experts from USDOT, state DOTs, Austria's ASFINAG, universities, associations and technology firms to examine how AI is being applied to building, operating and maintaining transportation systems. Sessions spanned safety, ITS and mobility, planning and project development, asset management, digital twins and data foundations.

The discussion stayed practical: what problem is being solved, what evidence shows AI solved it, and what it took organizationally to get there. The consensus was clear. The technology is now capable and accessible; the harder work is governance, data, workforce and change management.

14	43	100+	6
SESSIONS	SPEAKERS	ATTENDEES	THEMES

> The technology part of AI deployment is only 20% of the effort; the remaining 80% involves organizational and process change and workforce transformation.
>
> SYMPOSIUM CONSENSUS · THEME 4

THE BOTTOM LINE FOR LEADERS

01 AI is already delivering measurable results in safety, operations and inspection.

02 Results hinge on governed data and capable staff, not on which model is chosen.

03 Shared federal, pooled-fund and open-source resources mean no agency starts alone.

Six themes that ran through both days

Presentations and discussions converged on six themes. Each is paired below with the practical step speakers urged agencies to take.

01 Focus on the "why" for AI

Start with a real business problem and confirm AI is the right tool. Prove ROI in pilots before scaling, and learn to communicate that value to leadership.

ACT → **Define the problem, outcome and accountable owner before selecting a tool.**

02 Leverage the new capabilities AI enables

Computing power plus AI makes previously undiscoverable information usable — in imagery, data streams and documents — helping agencies become proactive, not reactive.

ACT → **Aim AI at bottlenecks where buried information slows decisions.**

03 Solidify the data foundation

Value from AI depends on well-organized, governed data. Integration and interoperability efforts are advancing within agencies, nationally and across state borders.

ACT → **Pair every AI strategy with a data strategy; let use cases expose the gaps.**

04 Prepare the organization and the workforce

Technology is about 20% of the effort; organizational and workforce change is the other 80%. Keep humans "at the helm" and address staff concerns openly.

ACT → **Fund AI literacy for every role and engage skeptics, not just enthusiasts.**

05 Keep pace with rapid AI development

AI capability is outpacing governance, procurement and institutional adoption. Don't "boil the ocean" — run small pilots with guardrails and stay agile.

ACT → **Set durable goals but keep technology choices flexible.**

06 Share information and tools

With pilots and studies multiplying, agencies need frequent collaboration and better channels to transfer practices and tools instead of reinventing them.

ACT → **Join pooled-fund efforts and reuse proven tools before building new ones.**

A PHRASE WORTH ADOPTING

Move beyond "human in the loop" to **"human at the helm"**: people remain responsible for reviewing, accepting and acting on every AI output.

What's working: evidence from the field

Selected results and lessons shared by presenters, by application area. Figures are as reported at the symposium; the full proceedings give context for each.

01
Safety
Reactive to predictive

- A USDOT rapid prototype fusing Waze, crash, weather and roadway data flagged high-risk corridors **12–22× better** than a naïve baseline.
- CTDOT and UConn measured near misses with **sub-$100 cameras**; the proof of concept led to a $1.57M Safe Streets for All grant.

02
Operations & ITS
Intervening earlier

- AI adaptive signals in Maricopa County, AZ cut vehicle intersection delay from **29 to 13 seconds** in a one-week evaluation.
- MIT's bus-headway model at CTA cut excess wait **~30%** with no added buses, though drivers followed it only ~40% of the time.

03
Planning & project development
Inventories at scale

- CTDOT's PRISM built a statewide pavement-marking inventory from imagery and LiDAR at **~97–98% accuracy**, tied to cost estimates.
- USDOT's AITPD program found agencies want AI tools that **fit existing workflows**, not raw data or stand-alone software.

04
Asset management & inspection
Continuous attention

- Reviewing years of reports and photos, a RIDOT pilot AI **independently flagged a bridge** that RIDOT had closed the week before.
- NJDOT's locally hosted bridge-inspection assistant cut analysis of 300+ page reports from **hours to minutes**.

05
Data foundations
From data to decisions

- UDOT pairs a bronze-to-platinum medallion architecture with a data mesh, aiming for **governance maturity Level 4**.
- PennDOT's **open-source Asset Fox** turns asset data into traceable treatment and project recommendations that staff review.

06
Knowledge & administration
Finding what you know

- ASFINAG's assistants have answered **50,000+ questions** (Agnes) and serve 200,000+ requests a year at about 3¢ each (Adam).
- UDOT's policy chatbot showed wrong answers often traced to **missing or duplicate documents**, not the model.

WATCH-OUTS SPEAKERS RAISED

A "99% accurate" model can still fail where it matters most · validate models locally and monitor for drift · avoid alert fatigue · prediction is not prevention.

The path forward: what agencies can do now

EIGHT MOVES FOR AGENCY LEADERS

01 **Start with the "why."** Name the problem, the outcome and an accountable owner before choosing any tool.

02 **Pilot small, then scale.** Document results, training and upkeep needs; expand only what proves its value.

03 **Fix data as you go.** Let each use case expose the governance, versioning and sharing gaps that actually matter.

04 **Govern by risk and AI type.** Machine learning, computer vision and generative AI differ in maturity; validate safety uses rigorously.

05 **Keep humans at the helm.** Staff own every decision and need clear thresholds for acting on an AI output.

06 **Invest in people and change.** Build AI literacy for every role, appoint champions and invite skeptics in.

07 **Budget the full lifecycle.** Plan for drift monitoring, retraining and usage (token) costs like any asset's O&M.

08 **Make the ROI case.** Track business outcomes, user benefits and augmentation — and the cost of not adopting.

RESOURCES TO WATCH

FHWA AI Deployment Center of Excellence
National hub for tools, best practices and proven use cases.

National AI Roadmap & Portal
Planned five-year roadmap and one entry point to vetted resources.

National AI Transportation Coalition
Pooled fund study to accelerate trustworthy AI adoption.

State DOT Council for Strategic AI Adoption
Pooled fund TPF-5(573), led by NJDOT.

ITS JPO Rapid Prototyping & Maturity Model
Low-risk pilots with USDOT and a readiness self-assessment.

ITS America AI Implementation Guide
Practical next steps plus a maturity self-assessment worksheet.

BEFORE ANY AI PROJECT, ASK

- ☐ What problem are we solving — and why AI?
- ☐ What is the risk if the output is wrong?
- ☐ How will we measure value and drift?
- ☐ Is our source data authoritative?
- ☐ Who is accountable for acting on it?
- ☐ Has someone already built this?

Go deeper. The full proceedings include session-by-session takeaways, the speaker roster and links to every initiative cited.

Program & materials: **www.ridotaisymposium.org**